2.2 Solving Inequalities Using Addition or Subtraction

Name _____ Period ____ Date _____

Algebra I 2.2 · Student sheet

I can…

- solve inequalities using addition.
- solve inequalities using subtraction and graph the solutions.
- write and solve inequalities for real-life problems.

Vocabulary · fill in the missing word

Addition Property of Inequality — _____ the same number to both sides keeps the inequality true

Subtraction Property of Inequality — subtracting the same number from both sides _____ the inequality true

Equivalent inequalities — inequalities with the same _____

Part 1 · Test the boundary

Inequality	Solution	Test 1st value	Test 2nd value	Test 3rd value
$x + 5 < 12$		$x = 6$:	$x = 7$:	$x = 8$:
$y - 3 \geq -1$		$y = 1$:	$y = 2$:	$y = 3$:

Did adding or subtracting ever change the direction of the symbol? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Addition and Subtraction Properties: Adding or subtracting the same number on both sides _____ change the symbol.

Check: Check one value _____ the solution and one _____ it.

Example 1

Solve $x - 6 \geq -10$.

Reading it backward: $4 \geq x$ means the same as x ____ 4.

Example 2

Solve $9 < y + 13$ and graph the solution.

Example 3

A number plus 8 is at most 3. Write and solve an inequality.

Example 4

You have $35 in your account and must keep at least $10 in it. How much can you spend?

Part 3 · Partner practice

Partner 1

Solve $m + 4.5 < 2$.

Partner 2

Solve $-3 \geq x - 7$ and graph.